EV Charging VR Research Audit

Oct 2, 2026 · @Naman Khatak

A grounded audit of the EV-charging cybersecurity VR project (BeamNG.tech + Varjo XR-4 + Muse S Athena): what the codebase and data actually show versus what the mid-sem report and literature review claim, and a research plan built on that evidence.

Project understanding

The project (MTP-1, IIT Mandi, Arpita Kumari + Naman Khatak, supervisor Prof. Varun Dutt, mentor Ranik Goyal) studies how people make decisions at compromised EV charging kiosks inside a VR driving simulation, and whether behaviour, gaze and EEG can flag the risky ones. The participant drives an EV in BeamNG.tech with a 400-point budget, reaches a station of three kiosks, and can charge, skip, or ask an in-sim AI assistant for help. Some kiosks are malicious: a fake discount coupon, incomplete charge delivery, or overcharging. Eye movements are meant to be captured with a Varjo XR-4 (built-in eye tracking) and EEG with a 4-channel Muse S Athena.

The mentor's four original objectives were refined mid-semester into: (1) push simulation FPS up from ~40-50; (2) build a multimodal deep-learning pipeline over an existing 120+-participant dataset that drives AI warnings; (3) collect new data to validate the warning mechanism; (4) a literature review and gap analysis for this new approach. Scenario expansion (more attack types) moved to future scope, with the design work already done.

This audit reads differently from the mid-sem report in one respect: the report and its supporting literature review describe the plan and intentions well, but this audit also opened the actual BeamNG.tech installation, the Python backend, and the 140 session folders already on disk, to check what is really implemented and what the existing dataset really contains - not just what the documents say it contains. Several differences surfaced; they're called out explicitly in Objectives 1-3 below, each marked by what evidence supports it.

Objective status

Treating every objective as uncompleted unless the code or the data on disk shows otherwise.

Objective	Current status	Required work	Evidence
1. FPS optimization	Some application-layer polling found; engine source is closed (compiled Bin64 + asset-only gameengine.zip) - no baseline FPS measurement exists in the repo	Establish a reproducible baseline, profile CPU/GPU/physics split, apply config-level and IPC-cadence fixes	Verified: Bin64 is compiled .exe/.dll; game-settings-default.cs has live shadow/reflection prefs; receive_ui.py has two 5 Hz polling loops
2. AI pipeline + warnings	A screenshot + RAG text-chat assistant is implemented and running; the eye-tracking/EEG-driven risk model and AI warning described in the report do not exist in code	Build the actual synchronization, feature extraction, and risk-classification layers; today's "AI" is a different system (deepfake-coupon detection chat), not the warning pipeline	Verified via ai_assistant.py, screenshot_analyzers.py, rag.py, receive_ui.py
3. New data collection	Not started (correct per report)	Define schema, synchronization and QC before collecting	Verified: no post-mid-sem session folders beyond the 140 already present
4. Literature review & gap analysis	Substantially complete and unusually rigorous - stronger than a typical mid-sem deliverable	Verify a handful of flagged citations (marked [to verify] in the report itself); extend to the corrected data picture	Verified by reading the review document in full - see Objective 4 below

The most consequential correction is inside Objective 2: the mid-sem report's Table 6 lists video, eye-tracking, EEG and event logs as available for all 120+ participants. The raw data on disk tells a different story - detailed in Objective 2.

Objective 1 - FPS optimization

The honest first finding: BeamNG.tech v0.38.3.0 is a closed commercial product here, not an open codebase. `Bin64/` holds compiled `.exe` / `.dll` files (BeamNG.tech.x64.exe, Vulkan layers, Ultralight), and `gameengine.zip` is assets only (cubemaps, datablocks, materials) - no C++ source. The render loop, physics stepper and VR compositor that the original brief asks for line-numbered fixes in are **not present in this install** and can't be profiled by reading code. What IS inspectable and genuinely "ours": the Lua gameplay layer, the custom `EVChargingGame` BeamNG UI app, the Python backend that drives it via BeamNGpy, and the TorqueScript graphics preferences. Below is what that layer actually shows, each item graded by how confident the evidence is.

Architecture (confirmed by reading the code, not assumed):

Varjo XR-4 (VR display + eye tracker, not yet wired to the pipeline) -> BeamNG.tech engine (closed-source renderer + soft-body physics) -> `evProjectBridge.lua` GE extension (event-driven message bridge) -> `receive_ui.py` Python backend (BeamNGpy client, owns game state) -> `EVChargingGame` AngularJS UI app (`app.js` / `app.html` , runs in the CEF overlay) -> per-session CSV logs under `backend/raw_data/` .

Bottleneck candidates found, with exact locations:

Priority	File	Function / lines	What it does	Expected impact	Confidence
Quick win	`settings/game-settings-default.cs:31-32`	`$pref::Shadows::disable` , `$pref::Shadows::textureScalar`	Shadows run at full resolution/quality by default (disable=0, textureScalar=1.0)	Medium-High	Config read directly; effect unmeasured
Quick win	`settings/game-settings-default.cs:22-24`	`$pref::Reflect::refractTexScale` , `frameLimitMS`	Reflection fill-rate budget is uncapped (frameLimitMS=0)	Medium	Config read directly; effect unmeasured
Quick win	`settings/game-settings-default.cs:38-42`	`$pref::BeamNGVehicle::dynamicReflection::*`	Dynamic vehicle reflections are already disabled by default - not a lever here	None (already off)	Verified
Medium	`backend/.../receive_ui.py:879-933` (`_ui_listener`)	`time.sleep(0.2)` loop issuing `queue_lua_command(..., response=True)` 5x/sec to read the UI bridge	Adds a synchronous Python<->Lua round trip every 200ms on a background thread	Low-Medium	Verified present; BeamNGpy's blocking-response semantics on the GE thread are UNVERIFIED without profiling
Medium	`backend/.../receive_ui.py:817-855` (`_odometer_worker`)	Same 5 Hz cadence, calls `vehicle.poll_sensors()` each tick	Second concurrent socket RPC stream at 5 Hz	Low-Medium	Verified present

Priority	File	Function / lines	What it does	Expected impact	Confidence
Needs verification	VR compositor / Vulkan rendering, soft-body physics step	Not accessible (compiled binary)	Most likely the dominant cost given BeamNG's known architecture (CPU-bound soft-body physics + stereo VR rendering)	Very High	UNVERIFIED - requires running the profiler-equipped build on the actual Windows workstation

The Varjo XR-4 panel refreshes at 90 Hz, so a stable 90 FPS is the correct, sufficient target - frames above the headset's refresh rate are never displayed. The report's stated 120 FPS target exceeds what the hardware can show and should be reframed as "stable 90, no drops," which is also the more defensible number to put in front of a reviewer.

Profiling methodology (since no baseline exists yet): run BeamNG.tech's built-in profiler (`gfxFrameOverlay` /in-engine FPS graph, or external tools - RenderDoc for a GPU frame capture, Windows Performance Analyzer for CPU) during a standard station-to-station drive, with and without the UI app and Python backend attached, to separate (a) base engine cost, (b) cost added by the custom scene content (kiosk props, station count), and (c) cost added by the IPC layer above. Without that split, any specific "shadows vs. physics vs. IPC" claim beyond what's in the table is a guess, not a finding - and the original brief's request for exact engine-level line numbers can't be met from this install regardless of effort spent re-reading it.

Objective 2 - AI / deep-learning pipeline

What is actually implemented today (verified by reading `ai_assistant.py`, `screenshot_analyzers.py`, `rag.py`, `aoi_mapper.py`, `receive_ui.py`): an in-game AI assistant the participant can pay 10 points to consult. On each station it takes a screenshot, runs it through an analyzer (likely a VLM call, given `screenshot_analyzers.py` + the `images/real/` vs `images/deepfake/` folders - this looks like deepfake-coupon detection, not gaze/EEG risk scoring), retrieves context with a small RAG index (`rag_index/index.faiss`), and returns a text recommendation plus TTS playback (`tts_player.py` , `audio_direction.py`). This is a real, working system - but it is a *different* system from the one described in the mid-sem report and script: a screenshot-analysis chatbot, not the "behavioural model decides risk -> warning" pipeline built on gaze + EEG. `aoi_mapper.py` exists (area-of-interest mapping groundwork) but nothing in `receive_ui.py` calls it yet. Status: **in progress, and currently on a different architecture than the one in the report** - this needs reconciling with the mentor before more pipeline work is built on either version.

What the existing dataset actually contains (verified by inspecting all 140 session folders in `backend/raw_data/`, not just trusting the report's Table 6):

Modality	Report claims (Table 6)	What's actually on disk	Confidence
Behavioural / interaction logs	Available, 120+	Present for all 140 sessions (`<id>.csv` interaction log + `demographics.json` + `consent.json` + `system_config.json`)	Verified
Eye tracking (gaze)	Available, 120+	Present for all 140 sessions - the large per-session `<id>.csv` file (up to ~193k rows / 80+ MB) is actually Varjo gaze output (columns: gaze_forward_x/y/z, pupil diameter, eye openness, etc.), not a summary log as the filename suggests	Verified, and a genuinely good finding - gaze coverage is better than the report implies
EEG (Muse S Athena)	Available, 120+	**Zero EEG files found anywhere in raw_data.** `backend/muse.py` is a standalone LSL recorder script, never imported by `receive_ui.py`, with no session-ID linkage	Verified absence - needs confirming with the team whether EEG lives elsewhere
Gameplay video	Available, 120+	**No video files found anywhere in raw_data.** The system instead captures single screenshots per station for the AI assistant	Verified absence
Data quality	Not yet assessed (per report, correctly)	One session is pre-labelled `session_..._126_glitched data` - a concrete, available example for the quality-check step	Verified

This changes Objective 2's scope: the "multimodal" pipeline currently has one reliable modality (gaze) plus behavioural logs, not three. EEG and video should be labelled **proposed for future collection**, not **existing data to analyze**, until the team locates EEG

recordings or confirms none were kept with session IDs.

Data schema (reflecting what's verifiably available, not the idealized version):

```
Participant ID, Session folder, Trial/Station number, Kiosk chosen, Attack
type (Safe/Cyber Attack),
Attack mode (blocked/reduced), Discount coupon shown, Decision
(charge/skip/AI-help), Points lost,
Decision latency, AI-assistant used (bool), Gaze stream (raw_timestamp,
gaze_forward_xyz, pupil diameter,
eye_openness) - present for ~140/140, EEG stream - present for 0/140
verified, Outcome label (risky/safe,
from game logs only, never from gaze)
```

Model plan, unchanged from the report's reasoning and still the right call given n~140: MLP baseline on fused features, then LSTM/GRU for the temporal decision sequence, a multimodal Transformer only if data quality supports it, each phase required to beat simple baselines (logistic regression, random forest, XGBoost) on participant-level held-out splits (GroupKFold / leave-participants-out). With EEG currently unverified as existing data, the first real pipeline iteration should be built and evaluated on gaze + behaviour alone, with EEG added later only if the team locates or newly collects synchronized recordings - this is a scope reduction, not a setback, and makes the n~140 overfitting risk the report already flags even more important to respect.

Evaluation: balanced accuracy / macro-F1 / AUROC (risky decisions are the minority class), an ablation across gaze-only vs. behaviour-only vs. combined, and participant-level cross-validation throughout - never a random trial-level split, which would let the model memorize individuals.

Objective 3 - data collection requirements

You're handling the physical collection yourselves, so this is deliberately just the spec the next round needs to produce scientifically usable data - not a logistics plan.

- **Fix the EEG gap first.** `muse.py` must write session-ID-tagged output into the same per-participant folder the gaze/behaviour logs use, on the same clock. Today it is a disconnected script - wiring it in is a prerequisite for any EEG analysis, not a data-analysis task.

- **Synchronization**: a shared timestamp or event marker across the gaze stream, the EEG stream, and the `interactions_*.csv` log, anchored on identifiable events already logged (`Flowgraph Reached` , `StationGenerated` , charging start/end) - `receive_ui.py` already timestamps every incoming/outgoing message (`_log_interaction`), so the event-marker backbone exists; it just needs the EEG stream joined to it.
- **Event markers already available in the logs**: `Started` , `Reached` (station N), `AudioTrigger` , `DrivingPageReached` , `ChargeUpdate` , `OutOfFuel` , `Finished` - these are the natural segmentation points for "approach -> notice -> decision -> response" windows.
- **Minimum required per session**: consent + demographics (already standard), gaze CSV, EEG CSV (once wired), interaction CSV, a system_config snapshot (already standard), and a pass/fail quality flag per modality rather than a single session-level pass/fail - the `126_glitched data` session shows a session can be fine on some modalities and bad on others.
- **Quality-control checks before a session counts toward the usable n**: gaze tracking-status column not predominantly invalid, EEG contact quality above threshold, event-marker count matching expected station count (should be ~20 `Reached` events per session), no duplicate/corrupted session folders (the repo already has at least one: `session_20260428_140404_032` has both `032.csv` and `032_1.csv`).

Objective 4 - literature review synthesis

The literature review document already provided ("Literature Review & Research-Gap Analysis") is unusually rigorous for this stage - sharper and more self-critical than most mid-sem reviews, and this audit builds on it rather than re-running the search. It covers seven domains (EV cybersecurity, human factors, eye tracking, EEG, EEG+eye fusion, VR/driving-simulator cybersecurity, AI trust/automation bias) with a structured evidence matrix (paper, participants, modality, finding, limitation) rather than a loose bibliography, and it distinguishes peer-reviewed sources (Nasr et al. 2022, Thomopoulos et al. 2024, McAlaney & Hills 2020, Horowitz & Kahn 2024) from items it itself flags `[to verify]` (the 2023 driving-sim ransomware/trust study, Soykan et al. 2021, Kurt et al. 2026's final DOI, the LSL citation) - that honesty about unverified citations is exactly the standard the original brief asks for, already applied.

Its most load-bearing, defensible claims: EV-charging cybersecurity research is almost entirely technical (Nasr et al.'s 16-system pentest, zero human subjects); phishing eye-tracking/EEG work is real but confined to static desktop screens; EEG+eye-tracking fusion research exists but is small and fragmented (Thomopoulos et al. screened 651 papers down to 18); and the single closest prior work, Kurt et al. (2026, maritime GNSS spoofing, 61 seafarers, EEG+eye-tracking), is a methodological twin but in a different domain, with passive detection rather than an active kiosk choice.

Two things this audit adds on top of it, both from the codebase/data check above:

- The review's EEG caution ("4-channel Muse, treat as exploratory/secondary") turns out to be more severe in practice than in theory - the current dataset has no EEG files at all, so the fusion literature it cites (gaze alone often matching or beating EEG+gaze fusion) is not just a risk to manage, it's close to the default outcome unless EEG gets collected and wired in.
- The review correctly privileges gaze as the primary modality; the data check confirms gaze coverage is actually better than the report elsewhere implies (all 140 sessions, not a small pilot), which strengthens rather than weakens the review's own recommended framing.

No new domain search is warranted at this stage - the gap is well-evidenced. What's worth doing next is verifying the four `[to verify]` citations before they appear in a manuscript, and folding the "deepfake-coupon screenshot assistant" architecture (Objective 2) into the literature frame, since it's closer to phishing/deepfake-detection UX research than to the gaze+EEG decision-making paradigm the rest of the review is built around.

Research gap

Carried over from the existing review (it is already evidence-based, not a manufactured "few studies combine X and Y" claim), with the data-reality check folded in.

Area	Existing research	Weakness of prior work	Opportunity here
EV cybersecurity	Extensive, technical (Nasr et al. pentested 16 real systems)	No human subjects anywhere in the technical literature	First human-in-the-loop EV-charging deception study

Area	Existing research	Weakness of prior work	Opportunity here
Human factors in security	Extensive	Almost entirely email/static-screen phishing	Interactive HMI deception + genuine EV charge-anxiety urgency
Eye tracking + security	Moderate (phishing)	Attention != accuracy; static AOIs only	Dynamic 3D VR AOIs, decision-linked gaze - and this project actually has gaze data for all 140 sessions to work with
EEG + security	Moderate	Needs many channels/trials for reliable signal	Feasible only as exploratory/secondary on a 4-channel Muse - and not yet collected here, which the research plan should say plainly
EEG + eye-tracking fusion	Small, fragmented (18 papers per Thomopoulos et al.)	Lab-static settings, small n	Answers the field's own call to extend into a realistic multi-sensory environment - once EEG exists
VR + cybersecurity	Exists	Almost entirely education/training, not measurement	A measurement paradigm instead of a training one
EV + human factors	Essentially absent	-	The core, well-evidenced gap
AI trust / automation bias	Robust behaviourally (Horowitz & Kahn, 9,000 participants)	No physiological measurement toward a *malicious* recommendation, anywhere, let alone in EV charging	A clean, cheap, within-subject manipulation (correct / absent / malicious AI advice) this project's existing assistant architecture can support with modest changes

The gap is genuine and well-supported. The risk is not the gap - it's claiming more coverage of it (EEG, video) than the current dataset actually delivers.

Novelty analysis

Candidate claim	Closest existing work	What's missing from it	Genuinely novel here?
"First to combine VR + EEG + eye tracking + AI"	Several partial combinations exist	N/A - this is a tool-count claim, not a contribution	No - avoid this framing entirely, as the review already insists
Human-in-the-loop study of interactive EV-charging deception, measured behaviourally + via gaze	Kurt et al. 2026 (maritime GNSS spoofing, EEG+eye, 61 seafarers, *passive detection*)	Different domain; passive detection, not an active multi-option choice	Yes - domain + the active-choice paradigm are both real differences
...with EEG as a confirmed modality	Kurt et al. 2026, Thomopoulos et al.'s 18-paper fusion corpus	Both have real EEG data; this project currently does not	Not yet - defensible only once EEG is actually collected and wired in; claiming it now would overstate the dataset
Automation bias toward a malicious (not just absent/correct) AI recommendation, with physiology	Horowitz & Kahn 2024 (behavioural, 9,000 participants, no physiology, not EV)	No physiological measurement of trust toward a malicious automated recommendation, in any domain found	Yes, and cheap to add - the existing AI-assistant architecture (Objective 2) is one condition-variable away from supporting correct / absent / malicious recommendation arms

Candidate claim	Closest existing work	What's missing from it	Genuinely novel here?
Deepfake/manipulated-image detection via an in-sim AI assistant	Not what any of the reviewed EV-security or phishing-eye-tracking literature studies	This is closer to a deepfake-detection-UX study than to the gaze+EEG risk paradigm	Possibly novel in its own right, but it's a *different* research question from the one in the mid-sem report, and should be named as such rather than folded in silently

The defensible core claim stays what the existing review already settled on: a human-in-the-loop investigation of decision-making under interactive EV-charging deception, in immersive VR, measured through behaviour and gaze (EEG and the automation-bias sub-study as stated extensions, not assumed ones).

Proposed research direction

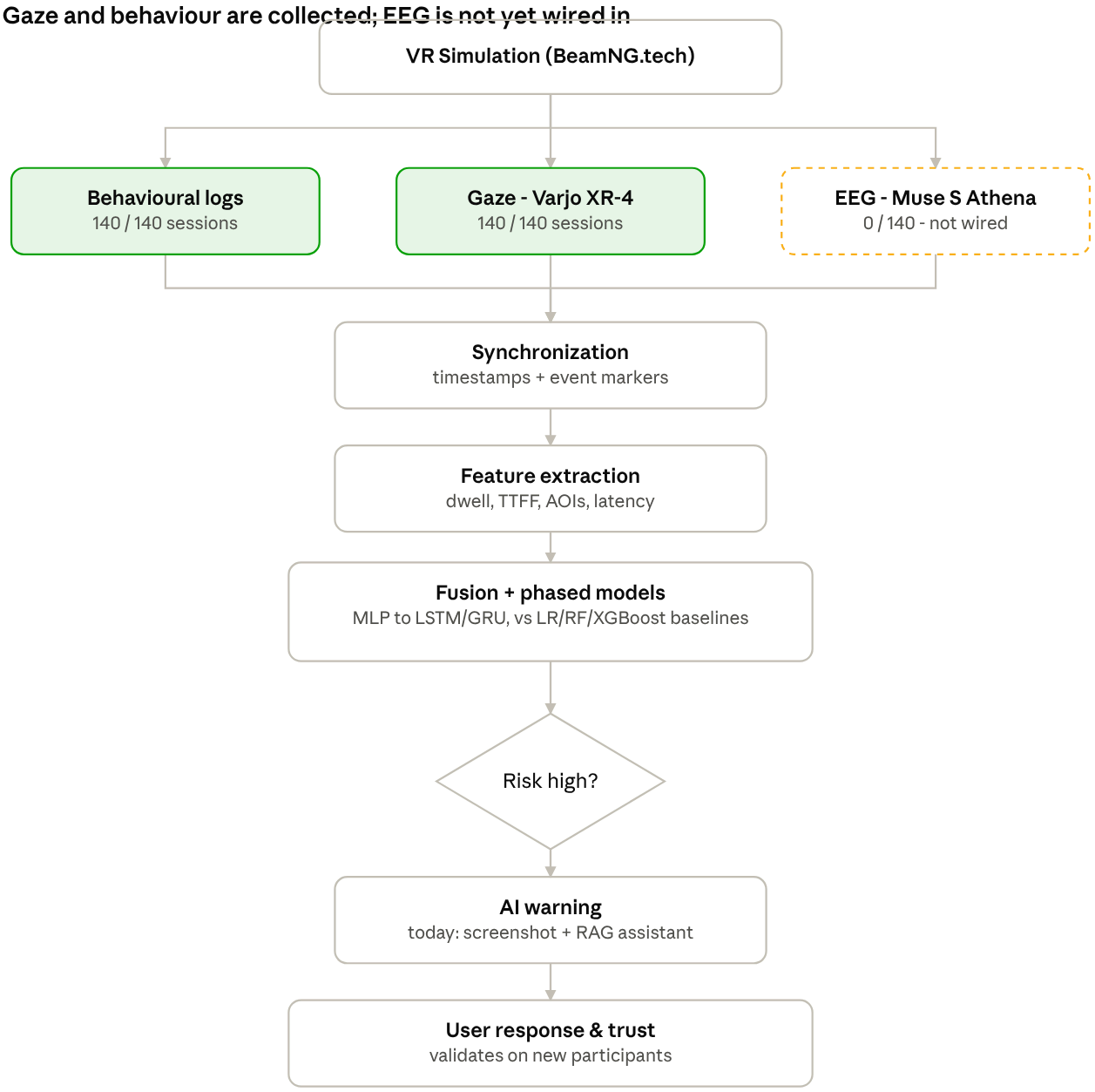
Primary research question: Does the presence and type of a simulated EV-charging cyberattack change decision outcomes and gaze-based attention patterns, and can those patterns, evaluated on held-out participants, flag risky decisions?

This is the review's own reframing of the report's un-testable "how can signals be used to identify vulnerability" question into one with a manipulable independent variable (attack present/absent, attack type) and measurable dependent variables (decision, latency, points lost, gaze dwell/TTFF on the coupon and price AOIs). It is chosen over the EEG-centred framing because it is the question the *existing* dataset (140 sessions, gaze + behaviour verified present) can actually answer today, without waiting on new EEG collection.

Secondary question (automation bias): Does a malicious AI-assistant recommendation increase unsafe kiosk choices relative to no recommendation or a correct one, and does independent verification behaviour (gaze to alternative kiosks/cues) mediate that effect? This reuses the existing AI-assistant code path (Objective 2) with one added condition variable, making it the cheapest genuinely novel extension available.

Explicitly deferred, not abandoned: any claim resting on EEG, until `muse.py` is wired into session logging and a pilot batch confirms usable signal quality in low-motion kiosk windows. Stating this as a phase gate (gaze+behaviour paper first, EEG-extended paper second) is more defensible to a reviewer than promising a three-modality study the current data can't yet support.

Complete research architecture



pipeline architecture - gaze and behaviour verified present for all 140 sessions, EEG not yet wired

The low-risk branch (not drawn) loops silently back to the next kiosk with no warning shown; only the high-risk branch reaches the AI-warning box. "User response & trust" at the bottom is the new-data-collection validation step, not a live feedback loop into the current session.

Experimental plan

Design: within-subjects - each participant already sees a mix of safe and attack kiosks across ~20 stations per session, so the existing 140-session dataset already has the structure a within-subject analysis needs; the automation-bias sub-study (correct / absent / malicious AI advice) should also be within-subject, randomized and counterbalanced across stations.

Sample size reasoning: n=140 (verified on disk) is large by the standards of comparable gaze/EEG human-factors studies (Kurt et al.: 61; VR driving-EEG studies: 15-31; phishing eye-tracking studies: 22-70) - more than adequate for mixed-effects behavioural/gaze analysis, but not large enough to justify a deep multimodal model as the primary analysis. Deep learning (MLP/LSTM/Transformer) should be positioned as a secondary, ablation-tested track behind the confirmatory statistics, exactly as the existing literature review recommends.

Statistics (primary, confirmatory): generalized linear mixed-effects models, e.g. `decision_outcome ~ attack_type * AI_recommendation_validity + (1|participant) + (1|station)`, which correctly treats the ~20 repeated decisions per participant as non-independent. Effect sizes with confidence intervals, not p-values alone; FDR correction across the many gaze features tested.

Labels: a decision is "risky" only from observable game-log outcomes (chose the compromised kiosk, accepted the fake coupon, lost points) - never inferred from gaze or EEG, which avoids the circular "EEG says risky, therefore risky" argument the review already flags.

Evaluation of any predictive model: participant-level GroupKFold / leave-participants-out cross-validation only; normalization and feature selection fit on training folds alone; report balanced accuracy, macro-F1 and AUROC (risky decisions are the minority class) against a majority-class baseline and a behaviour-only baseline.

What would count as failure, stated up front: if gaze features show no reliable difference between risky and safe decisions after FDR correction, or if a trained model cannot beat the behaviour-only baseline on held-out participants, that is a reportable negative result,

not a reason to keep searching for significance.

Implementation roadmap

Quick wins (days, minimal risk)

- Lower `Shadows::textureScalar` and raise `Shadows::disable` in `settings/game-settings-default.cs:30-32`; measure FPS delta before/after with the in-engine overlay.
- Cap `Reflect::frameLimitMS` (`game-settings-default.cs:23`) to bound reflection cost per frame.
- Reconcile the two divergent `main.py`/config copies (outer `backend/` vs. deployed `EVChargingGame/backend/`) - they currently differ on `starting_points`, `incident_mode`, and attack probabilities, which is a reproducibility risk for any analysis comparing sessions across dates.

Medium (1-2 weeks)

- Run a proper CPU/GPU/physics profiling pass (RenderDoc / Windows Performance Analyzer) to replace the UNVERIFIED engine-level guesses in Objective 1 with measured numbers.
- Wire `muse.py` into `receive_ui.py` with session-ID-tagged output on the shared event-marker timeline.
- Build the gaze-only + behaviour-only feature extraction and baseline models (logistic regression, random forest, XGBoost) on the existing 140-session dataset - this doesn't require new data collection or EEG.
- Reconcile the AI-assistant architecture (screenshot+RAG deepfake-detection chat) with the report's described gaze/EEG-driven warning pipeline; decide with the mentor which one the "AI warning" objective actually targets.

Deep (architectural, next phase)

- Full multimodal pipeline (gaze + EEG + behaviour) once EEG is verifiably collected and synchronized for a meaningful subset of new participants.
- AI-warning integration that actually consumes the trained risk model's output, with the correct/absent/malicious recommendation conditions for the automation-bias sub-study.

- New data collection round incorporating the Objective 3 schema and QC checks above.

Professor presentation summary

A one-page version of this audit's three load-bearing findings, for the next meeting with Prof. Dutt and Ranik Goyal.

1. **The dataset is stronger on gaze than the mid-sem report suggests, and has no EEG yet.** All 140 session folders (not 120+) contain Varjo gaze data at full fidelity - this was verified by inspecting the files, not assumed. No session contains EEG or video, despite Table 6 of the report listing both as available. `muse.py` is a disconnected script. This should be corrected before the next report draft, and reframes the pipeline as gaze+behaviour now, EEG as a stated future extension.
2. **The in-sim "AI assistant" that exists in code today is a screenshot + RAG deepfake-detection chatbot, not the gaze/EEG-driven risk-warning system described in the script and report.** Both are legitimate research directions, but they are different systems, and the team should explicitly decide with the mentor which one "AI warning" refers to going forward.
3. **FPS optimization is bounded by what's inspectable.** BeamNG.tech's renderer and physics engine are closed-source in this install (compiled binaries + asset-only zip) - exact engine-level line numbers, as the original task brief asked for, aren't obtainable from this codebase. What is real and actionable: two concrete graphics-config quick wins (shadow/reflection settings) and two IPC polling loops in the Python backend running at 5 Hz each - worth testing, but neither is confirmed as the dominant cost without an actual profiling run on the Windows workstation. The 120 FPS target should be revised to a stable 90, matching the XR-4's refresh rate.

The literature review and gap analysis (Objective 4) remain the project's strongest asset as-is - no changes recommended beyond verifying the four citations it already flags `[to verify]`.